Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

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I. WORK DONE THIS WEEK

In Week 4, we established a measurable baseline for fabric defect detection using the AITEX Fabric Image Database. The selected dark-contrast method achieved a pooled pixel-level F1-score of 0.0316 and an intersection over union (IoU) of 0.0161 in the Week 5 notebook's re-evaluation. However, the earlier experiments showed that detecting only dark regions is not sufficient for different types of fabric defects. This week, we therefore focused on investigating multiple anomaly cues, combining their scores, and evaluating the resulting methods using the same held-out testing protocol.

A. A. Dataset and Experimental Setup

We continued using the AITEX Fabric Image Database (AFID) because it provides defective fabric images with corresponding ground-truth masks. The dataset inspection identified 106 defective-image entries, 107 mask files, and 141 defect-free images. After checking image-mask correspondence and dimensions, 105 valid image-mask pairs were retained for evaluation. One entry had no valid matching mask.

We maintained the reproducible image-level split using a random seed of 42. The 105 valid pairs were divided into 84 training images and 21 held-out test images. An assertion confirmed that no image was shared between the two sets. Since some defect codes contained very few images, stratification was not feasible.

The feature maps and thresholds were developed using the training set. The selected thresholds were then kept fixed during held-out testing. This was important for avoiding the use of test-set results when choosing the final detector configuration.

TABLE I
DATASET SUMMARY FOR WEEK 5

Item	Count
Defective-image entries	106
Ground-truth mask files	107
Defect-free images	141
Valid image-mask pairs	105
Training images	84
Held-out test images	21
Train/test overlap	0

B. B. Investigation of Multiple Anomaly Cues

We implemented four patch-level anomaly cues to examine different characteristics of fabric defects. Each image was divided into non-overlapping 16×16 patches. For an image of size 256×4096 pixels, this produced a grid of 16×256, or 4096 patches.

The four cues were:

- **Dark contrast:** Measures how much darker a patch is than the median intensity of its neighbouring patches.
- **Bidirectional contrast:** Considers intensity deviations in both directions, allowing potentially dark and bright anomalies to be investigated.
- **Texture deviation:** Measures local differences in texture features relative to neighbouring patches.
- **Gradient deviation:** Examines changes in gradient information that may indicate irregular edges or local structural changes.

These cues were intended to capture complementary characteristics rather than relying entirely on the intensity of a patch. We compared their anomaly maps and measured their ability to distinguish defect-positive patches from normal patches.

On the sample image, dark contrast achieved a ROC-AUC of 0.9753, while bidirectional contrast achieved 0.9487. Texture deviation and gradient deviation achieved ROC-AUC values of 0.7668 and 0.7843, respectively. The average precision values were 0.6546, 0.6184, 0.0323, and 0.0448, respectively. These results suggested that intensity contrast was particularly useful for the examined sample, although they did not establish that the same ranking would hold across the entire dataset.

C. C. Weighted Multi-Cue Fusion

We next investigated whether combining different anomaly cues could improve defect detection. Equal-weight fusion was compared with a contrast-dominant fusion method. The original dark-contrast detector was retained as the baseline.

The sample-image comparison gave the following results:

The contrast-dominant method obtained a slightly higher ROC-AUC than the dark-contrast baseline on this sample. However, equal-weight fusion performed worse on both reported measures. This demonstrated that combining features does not automatically improve performance; the contribution of each cue must be evaluated carefully.

TABLE II
SAMPLE-IMAGE FUSION COMPARISON

Method	ROC-AUC	Average Precision
Dark contrast baseline	0.9753	0.6293
Equal-weight fusion	0.9633	0.4702
Contrast-dominant fusion	0.9759	0.5897

D. D. Threshold Selection and Held-Out Testing

We evaluated candidate thresholds using training data only. Percentile-based thresholds were investigated for the dark-contrast, bidirectional-contrast, and contrast-dominant fusion methods. The selected thresholds were frozen before the 21 held-out images were evaluated.

The selected configurations were:

TABLE III
SELECTED TRAINING THRESHOLDS

Method	Percentile	Threshold
Dark contrast	98.5	3.8259
Bidirectional contrast	97.5	3.9433
Contrast-dominant fusion	97.5	2.8562

Table IV presents the pooled pixel-level results on the held-out test set. The predictions were expanded from patch decisions to pixel masks and compared with the ground-truth masks.

TABLE IV
HELD-OUT TEST RESULTS

Method	Prec.	Recall	F1	IoU
Dark contrast	0.0169	0.2535	0.0316	0.0161
Bidirectional contrast	0.0223	0.5606	0.0430	0.0220
Contrast-dominant fusion	0.0239	0.6467	0.0460	0.0235

The contrast-dominant method obtained the highest pooled precision, recall, F1-score, and IoU among these three configurations. Its pooled F1-score was approximately 45.6% higher than that of the Week 5 dark-contrast baseline. Nevertheless, its precision and IoU remained low, indicating that a large number of normal pixels were still incorrectly classified as defective.

E. E. Additional Spatial and Frequency-Domain Experiments

We also investigated spatial background suppression, local adaptive scoring, and Fourier-domain filtering. The spatial background suppression experiment obtained a ROC-AUC of 0.5396 and average precision of 0.0096 on the examined sample. Local adaptive scoring obtained a ROC-AUC of 0.8588 and average precision of 0.0202, which was substantially weaker than the dark-contrast baseline on that sample.

In the Fourier-domain experiment, we examined the dominant non-zero frequency components and implemented a corrected filtering method to attenuate selected frequency components. The corrected filter preserved the original image dimensions. However, quantitative evaluation showed that the filtering reduced the dark-contrast detector's sample-image ROC-AUC from 0.9753 to 0.9085 and average precision from 0.6293 to 0.2093. Therefore, the tested Fourier filtering configuration was not retained as an improvement to the current detection pipeline.

II. WORK TO BE DONE NEXT WEEK

Next week, we plan to focus on reducing false positives and improving the localization of small defects. The Week 5 experiments indicate that the contrast-dominant fusion method improves recall, but it still marks many normal fabric pixels as defective.

First, we will investigate more effective combinations of the intensity, texture, and gradient cues. We will examine whether different weights or a more selective combination can preserve the recall improvement while reducing false detections.

Second, we will investigate the effect of patch size and prediction refinement. Since a 16×16 patch is classified as a single unit, even a small defect can cause the entire patch to be marked as defective. We will explore ways to refine the predicted regions without losing the small defects that the detector is intended to identify.

Third, we will study threshold sensitivity and the possibility of using locally estimated background statistics. Any new threshold or parameter selection will be performed using training data, followed by evaluation on the held-out test set.

Finally, we will organize the experiment outputs, comparison tables, and visual results in the GitHub repository. We will also document the failure cases and prepare the next version of the weekly report.

III. LITERATURE REVIEW

This week, the literature review helped us interpret the behaviour of the different anomaly cues and understand the limitations of the current approach.

The work on modified Local Binary Patterns (LBP) [1] and GLCM-based fabric defect detection [2] provides the basis for representing local texture and statistical relationships between neighbouring gray levels. The combined use of LBP and GLCM [3] is also relevant to our experiments because the two descriptors can represent different aspects of fabric texture.

Our experiments indicate that texture deviation alone does not necessarily outperform intensity contrast. This observation motivates further investigation of feature and score fusion. The work on decision fusion for textile inspection [4] is relevant to this direction, as it considers combining information for visual inspection. Similarly, the comparison of GLCM and Gabor filters [5] provides a possible direction for future experiments if the present texture features remain insufficient.

The AITEX database [6] continues to be important for our evaluation because its ground-truth masks allow pixel-level precision, recall, F1-score, and IoU to be measured. This makes it possible to identify the difference between detecting anomalous regions and accurately localizing the actual defect pixels.

IV. CHALLENGES FACED

The first challenge was dealing with the extreme imbalance between defect and normal pixels. In the sample image examined during Week 5, only 0.133% of the pixels belonged to the ground-truth defect. Consequently, a detector can identify many defect pixels while still generating a large number of false positives.

The second challenge was selecting an effective combination of anomaly cues. Although contrast was strong on the examined sample, texture and gradient features behaved differently. Equal-weight fusion did not improve the sample-image results, showing that the cues cannot simply be combined without considering their relative usefulness.

Another challenge was the coarse spatial resolution of patch-based predictions. Expanding a patch-level decision to a complete 16×16 block can include many normal pixels around a small defect. This limits the attainable precision and IoU even when the correct region is partially detected.

Finally, the additional spatial and Fourier-domain experiments did not improve the sample-image metrics. The results show that more processing is not necessarily beneficial and that each modification must be evaluated quantitatively. The small held-out set and uneven distribution of defect codes also limit how confidently the current results can be generalized to every type of fabric defect.

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